

Finite-Dimensional Davies Interface Lemmas and TFIM Witness Tests for the Heisenberg Cut as a Resource Boundary

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Status of claims (read first). *Imported result:* the maintenance inequality, in the corrected form of [2] (v2) — unconditional entropy-production floor; efficient baselines; free-baseline case (Theorem 2.1). *Proved here (finite-dimensional):* the exact $\omega = 0$ Dirichlet identity (Lemma 4.1), the corrected modular-weighted Bohr-block identity (Lemma 4.3; v1’s commutator version was false for $\omega \neq 0$), and the witness lower bound (Proposition 4.2). *Sufficient conditional bound:* the linear envelope suppression lemma (Lemma 4.6) under explicit hypotheses; v1’s quadratic version is withdrawn as proved and reposed as an open question (Remark 4.7). *Numerics:* exact diagonalization, $N \leq 12$, stability-checked finite-size diagnostics; no thermodynamic-limit uniformity claimed. All scripts and logs in the companion repository (verification/0073).

Abstract

We present a finite-dimensional technical core for interpreting the quantum–classical boundary as a control-resource limitation: a state is operationally “classical” (relative to a chosen conditional expectation) whenever sustaining coherence is infeasible under a finite power budget. Using battery-assisted thermal operations at temperature T and energy pinching Δ , coherence is $C(\rho) = S(\rho \parallel \Delta[\rho])$ with instantaneous loss rate $\dot{C}_{\text{loss}}(\rho)$ under uncontrolled Markovian dynamics; the imported maintenance bounds of [2] (v2) give $P_{\text{extra}} \geq k_B T \dot{C}_{\text{loss}}$ in their corrected operational senses. We isolate the dynamical hinge required for geometric scaling claims and prove two interface results for Davies generators: a *negative* result — an active $\omega = 0$ channel yields a witness-based rate lower bound via a KMS commutator with $S(0)$ (exact identity, Lemma 4.1) — and a *corrected exact Bohr-block identity* for the KMS Dirichlet form with modular weights (Lemma 4.3), which replaces v1’s commutator decomposition (measurably false for $\omega \neq 0$: 30–40% deviation at $\beta = 1$, while the corrected form is machine-exact). A *positive* sufficient lemma gives linear envelope suppression under infrared exclusion, quasi-local Bohr spectral tails, and thermal clustering of the reference (Lemma 4.6); whether the Davies class supports the quadratic envelope — which *is* exact in the local-sink quasi-free class [3] — is stated as an open question with the obstruction quantified. TFIM exact-diagonalization witness protocols ($R_{\text{opt}}(\epsilon)$, scaling and temperature sweeps) and state-side micro-tests (plateau of $\dot{C}_{\text{loss}}/C$, linear $\gamma(0)$ and $1/\beta$ scalings, with step-size and PSD-clipping stability checks) are reported with reproducible conventions.

1 Scope, non-claims, and program map

This work does not propose modified quantum dynamics, collapse mechanisms, derivations of the Born rule, or ontological resolutions of the measurement problem. The claims are operational: minimal control resources required to maintain coherence against uncontrolled open-system dynamics under an explicit thermodynamic bookkeeping model.

ID	Paper	Role in the series (all at v2, 2026)
P1	ai.viXra:2512.0060	AQFT clustering; conditional-reattachment reconstruction (Petz misidentification corrected)
P2	ai.viXra:2512.0061	Maintenance work theorem (work sign and extra-power definition corrected in v2)
P3	ai.viXra:2512.0064	Heisenberg cut as resource boundary; exact rate-inheritance law (v1 windowed evidence with)
P4	ai.viXra:2512.0072	RIP framing note (weak form as lemma; strong-form status updated)
P5	ai.viXra:2512.0071	Architectural closure map
P6	ai.viXra:2512.0070	Davies-class stress test (ceiling/floor envelopes; secular delocalization)
P7	ai.viXra:2512.0073	This work: finite-dimensional Davies interface lemmas, TFIM witnesses, micro-tests

All papers, verification scripts, and reference logs are distributed in a single companion repository; this paper’s scripts live in `verification/0073`. (v1’s remark expecting loose figure files at compile time is gone: all numerical artifacts are script-generated.)

2 Finite-dimensional operational core

2.1 Thermal state, energy pinching, and coherence

Let $H_S = \sum_n E_n \Pi_n$ and, at temperature T , $\gamma_S := e^{-\beta H_S} / \text{Tr}(e^{-\beta H_S})$, $\beta = 1/(k_B T)$. Define $\Delta[\rho] := \sum_n \Pi_n \rho \Pi_n$ and $C(\rho) := S(\rho \| \Delta[\rho])$ (nats), $S(\rho \| \sigma) := \text{Tr}[\rho(\log \rho - \log \sigma)]$.

2.2 Uncontrolled dynamics and loss rate

For a Markovian semigroup $\rho_t = e^{t\mathcal{L}}(\rho)$ with $\mathcal{L}(\gamma_S) = 0$, define $\dot{C}_{\text{loss}}(\rho) := -\frac{d}{dt}C(\rho_t)|_{t=0}$.

2.3 Control model and imported maintenance bounds

Theorem 2.1 (Imported, corrected form; proved in [2] (v2)). *Under stroboscopic battery-assisted thermal operations at temperature T : (a) $P_{\min}(\rho) \geq k_B T \sigma(\rho)$ unconditionally, with $\sigma(\rho) \geq 0$ the entropy production rate, split as $\sigma = \dot{F}_{\text{diag}}^\Delta + \dot{C}_{\text{loss}}$; (b) $P_{\min}(\rho) \geq k_B T \dot{C}_{\text{loss}}(\rho)$ under diagonal contraction (automatic for Davies generators); (c) against thermodynamically ε -efficient population baselines, under population autonomy, $P(s) - P(s_{\text{diag}}) \geq k_B T \dot{C}_{\text{loss}}(\rho) - \varepsilon$; in the free-baseline case ($\mathcal{L}(\Delta[\rho]) = 0$ and $\Delta[\mathcal{L}(\rho)] = 0$, e.g. pure dephasing), $P_{\text{extra}}(\rho) = P_{\min}(\rho) \geq k_B T \dot{C}_{\text{loss}}(\rho)$ with no further assumptions.*

Remark 2.2 (Correction relative to v1). v1 imported this as “ P_{extra} , defined as an infimum over pairs of strategies, satisfies $P_{\text{extra}} \geq k_B T \dot{C}_{\text{loss}}$ whenever both control sets are non-empty”. That unlinked-pairs infimum is $-\infty$ (battery-burning baselines) and the v1 statement is vacuous; see [2] (v2) and its changelog. All uses below are through Theorem 2.1(a)–(c).

Corollary 2.3 (No-go from a uniform loss floor). *Fix $C_0 > 0$ and a nonempty family $\mathcal{F}_{C_0} \subset \{\rho : C(\rho) \geq C_0\}$ within the scope of Theorem 2.1(b) or (c). If $g(C_0) := \inf_{\rho \in \mathcal{F}_{C_0}} \dot{C}_{\text{loss}}(\rho) > 0$, then maintaining any $\rho \in \mathcal{F}_{C_0}$ requires power at least $k_B T g(C_0)$ (total power under (b); incremental over efficient baselines under (c)); if $P_{\max} < k_B T g(C_0)$, sustained maintenance is operationally impossible on $\mathcal{F}_{C_0}$.*

3 Davies generators, KMS Dirichlet form, and envelopes

3.1 Bohr components, inner product, Dirichlet form

For a single Hermitian coupling $S = S^\dagger$, define Bohr components $S(\omega) := \sum_{E_m - E_n = \omega} \Pi_m S \Pi_n$, so $S = \sum_\omega S(\omega)$ and $S(\omega)^\dagger = S(-\omega)$; bath rates obey KMS: $\gamma(-\omega) = e^{-\beta\omega} \gamma(\omega)$. With $\sigma = \gamma_S$, the KMS inner product is $\langle A, B \rangle_\sigma := \text{Tr}(\sigma^{1/2} A^\dagger \sigma^{1/2} B)$, and for the Heisenberg generator $\mathcal{L}^\dagger$ the Dirichlet form is $\mathcal{E}_\sigma(O) := -\text{Re} \langle O, \mathcal{L}^\dagger(O) \rangle_\sigma$.

3.2 Separation and envelopes (ceiling vs. floor)

Fix a coupling region (site j_0) and let $\mathcal{A}_\epsilon$ be operators supported at distance $\geq \epsilon$. Define $\kappa(\epsilon) := \sup_{O \in \mathcal{A}_\epsilon} \mathcal{E}_\sigma(O) / \|O\|_{2,\sigma}^2$.

Remark 3.1 (Ceiling annotation, inherited from [4] (v2)). $\kappa(\epsilon)$ is a *ceiling*; witness lower bounds (Proposition 4.2) bound it from below, which is the correct direction for *rate-floor* (negative) conclusions. Maintenance *cost* conclusions on state families use the sector quantities of Section 7, not κ ; the floor envelope $\kappa_\perp$ relevant there is developed in [4] (v2).

3.3 Pauli-string tail map

With Λ the sites and $\Omega_\epsilon := \{j : d(j, j_0) \geq \epsilon\}$, define $E_{<\epsilon}$ as the orthogonal projection (normalized Hilbert–Schmidt) onto Pauli strings supported in $\Lambda \setminus \Omega_\epsilon$, and $\text{tail}_\epsilon(X) := X - E_{<\epsilon}[X]$. Norm equivalences with $\|\cdot\|_{2,\sigma}$ hold at fixed (N, β) with constants containing $\lambda_{\min}(\sigma)^{-1/2}$ factors.

4 Interface lemmas

4.1 Negative lemma: the $\omega = 0$ channel (exact; unchanged)

Lemma 4.1 (Exact $\omega = 0$ Dirichlet identity). *If the $\omega = 0$ channel is $\mathcal{L}_0^\dagger(O) = \gamma(0)[S(0)OS(0) - \frac{1}{2}\{S(0)^2, O\}]$ with $S(0) := \sum_n \Pi_n S \Pi_n$ and $\gamma(0) > 0$, then for all O ,*

$$\mathcal{E}_\sigma^{(0)}(O) = \frac{\gamma(0)}{2} \| [S(0), O] \|_{2,\sigma}^2. \quad (1)$$

Proof. $S(0)$ commutes with H_S , hence with $\sigma^{1/2}$; expanding both sides of the KMS pairing and matching terms gives the identity (all modular weights are 1 at $\omega = 0$). $\square$

Proposition 4.2 (Fixed-separation witness lower bound). *If there exist $c_{\epsilon_0} > 0$ and $O_{\epsilon_0} \in \mathcal{A}_{\epsilon_0}$ with $\| [S(0), O_{\epsilon_0}] \|_{2,\sigma}^2 / \| O_{\epsilon_0} \|_{2,\sigma}^2 \geq c_{\epsilon_0}$, then $\kappa(\epsilon_0) \geq \frac{\gamma(0)}{2} c_{\epsilon_0}$.*

4.2 Corrected exact Bohr-block identity

Lemma 4.3 (Modular-weighted Bohr-block identity (corrected)). *For the single-channel Davies generator $\mathcal{L}^\dagger(O) = \sum_\omega \gamma(\omega)[S(\omega)^\dagger OS(\omega) - \frac{1}{2}\{S(\omega)^\dagger S(\omega), O\}]$ and Hermitian O ,*

$$\mathcal{E}_\sigma(O) = \sum_\omega \gamma(\omega) e^{\beta\omega/2} \text{Re} \left[\frac{1}{2} \| S(\omega) O \|_{2,\sigma}^2 + \frac{1}{2} \| OS(\omega)^\dagger \|_{2,\sigma}^2 - \langle S(\omega) O, OS(\omega) \rangle_\sigma \right]. \quad (2)$$

At $\omega = 0$ the bracket equals $\frac{1}{2} \| [S(0), O] \|_{2,\sigma}^2$, recovering Lemma 4.1.

Notation. In (2) the two norms involve $S(\omega)O$ and $OS(\omega)^\dagger$, while the cross term is $\langle S(\omega)O, OS(\omega) \rangle_\sigma$ — with $S(\omega)$, not $S(\omega)^\dagger$, in the second slot. The bracket is therefore *not* a commutator norm, and the expression does not reduce to v1’s commutator square except at $\omega = 0$ (where $S(0)^\dagger = S(0)$ and all three terms assemble into $\frac{1}{2} \| [S(0), O] \|_{2,\sigma}^2$).

Proof. Fix ω , write $L := S(\omega)$. Since L has definite Bohr frequency, $\sigma^{1/2}L = e^{-\beta\omega/2}L\sigma^{1/2}$ and $\sigma^{1/2}L^\dagger = e^{+\beta\omega/2}L^\dagger\sigma^{1/2}$. Then, for Hermitian O ,

$$\langle O, L^\dagger OL \rangle_\sigma = e^{\beta\omega/2} \langle LO, OL \rangle_\sigma, \quad (3)$$

$$\langle O, L^\dagger LO \rangle_\sigma = e^{\beta\omega/2} \langle LO, LO \rangle_\sigma, \quad (4)$$

$$\langle O, OL^\dagger L \rangle_\sigma = e^{\beta\omega/2} \langle OL^\dagger, OL^\dagger \rangle_\sigma, \quad (5)$$

by moving one $\sigma^{1/2}$ through $L^\dagger$ (first two) or cycling L through the trace (third). Assembling $-\text{Re}\langle O, \mathcal{L}_\omega^\dagger O \rangle_\sigma$ gives the ω -summand of (2). $\square$

Remark 4.4 (v1's Lemma 2 was false). v1 asserted $\mathcal{E}_\sigma(O) = \frac{1}{2} \sum_\omega \gamma(\omega) \| [S(\omega), O] \|_{2,\sigma}^2$. This omits the modular weights and mis-shapes the cross term; numerically (TFIM, $N = 8$, $\beta = 1$, random Hermitian O) it deviates from the true Dirichlet form by 30–40%, while (2) agrees to 2×10^{-10} (Appendix E). Everything in v1 that used only the $\omega = 0$ case (the witness program of Section 5) is unaffected.

4.3 Positive lemma: linear envelope suppression (corrected scope)

Definition 4.5 (Infrared exclusion). The coupling satisfies IR exclusion at scale $\omega_* > 0$ if $S(\omega) = 0$ for all $0 < |\omega| < \omega_*$ and the $\omega = 0$ channel is absent ($\gamma(0) = 0$ or $S(0) \in \mathbb{C}\mathbf{1}$).

Lemma 4.6 (Linear envelope suppression (sufficient)). *Assume: (i) IR exclusion; (ii) quasi-local Bohr tails: there are $A, \xi > 0$, $p \geq 0$ with $\|\text{tail}_\epsilon(S(\omega))\|_\infty \leq A(1 + \epsilon)^p e^{-\epsilon/\xi}$ for all $|\omega| \geq \omega_*$; (iii) bounded weighted rates: $\Gamma_\beta := \sum_{|\omega| \geq \omega_*} \gamma(\omega) e^{\beta\omega/2} < \infty$; (iv) thermal clustering across the cut: for $O \in \mathcal{A}_\epsilon$ and near-supported P , the KMS-norm factorization defects obey $|\|OP^\dagger\|_{2,\sigma}^2 - \|OP\|_{2,\sigma}^2| \leq B e^{-\epsilon/\xi'} \|P\|_\infty^2 \|O\|_{2,\sigma}^2$ for some $B, \xi' > 0$. Then there is $C > 0$ (explicit in $A, B, \Gamma_\beta, \|S\|_\infty$) with*

$$\kappa(\epsilon) \leq C(1 + \epsilon)^p e^{-\epsilon/\min(\xi, \xi')}. \quad (6)$$

All constants are fixed- (N, β) quantities unless hypotheses (ii)–(iv) are assumed uniformly: Γ_β , the norm-equivalence factors, and the clustering constants of (iv) may all depend on size and temperature, and no thermodynamic-limit uniformity is claimed.

Proof sketch. Apply Lemma 4.3. For $O \in \mathcal{A}_\epsilon$ split each block $S(\omega) = P_\omega + T_\omega$ with $P_\omega := E_{<\epsilon}[S(\omega)]$ (near-supported, hence commuting with O and with $O^\dagger$) and $T_\omega := \text{tail}_\epsilon(S(\omega))$. Substituting into (2) and using $[P_\omega, O] = [P_\omega^\dagger, O] = 0$, every term containing at least one T_ω is bounded via Cauchy–Schwarz by $\|T_\omega\|_\infty (\|S\|_\infty + \|T_\omega\|_\infty) \|O\|_{2,\sigma}^2$; the T -independent residue of each block is $\frac{1}{2}(\|OP_\omega^\dagger\|_{2,\sigma}^2 - \|OP_\omega\|_{2,\sigma}^2)$, which hypothesis (iv) bounds by $B e^{-\epsilon/\xi'} \|S\|_\infty^2 \|O\|_{2,\sigma}^2$ (splitting defects between P_ω and $S(\omega)$ are again T -linear). Summing with weights $\gamma(\omega) e^{\beta\omega/2} \leq \Gamma_\beta$ and taking the supremum over O gives the claim. $\square$

Remark 4.7 (The quadratic envelope is withdrawn as proved, and reposed). v1 claimed $\kappa(\epsilon) \leq C(1 + \epsilon)^{2p} e^{-2\epsilon/\xi}$ (tail-squared). Its proof used the false identity of Remark 4.4, and with the corrected identity the splitting produces T -linear cross terms and T -independent thermal-clustering residues that the commutator cancellation of v1 no longer removes. An exact “tail-reduction” shortcut ($\mathcal{E}_\sigma(O; \{S(\omega)\}) = \mathcal{E}_\sigma(O; \{T_\omega\})$) is also numerically false — measured relative defect $\approx 5 \times 10^{-3}$ at $(N, \beta, \epsilon) = (8, 1, 3)$ (Appendix E) — because the near parts P_ω do not have definite Bohr frequency, so the pair-positivity argument that would kill the cross terms does not apply. Whether the Davies class supports a quadratic envelope under strengthened hypotheses is open; note that the quadratic (amplitude-squared) exponent *is* exact in the local-sink quasi-free class, $\kappa(\epsilon) \propto e^{-2q(\omega_b)\epsilon}$ [3], and the distinction between correlator-mediated (linear) and amplitude-mediated (squared) envelopes is precisely the one drawn in [6] (v2).

5 TFIM witness computations (exact diagonalization)

5.1 Model and conventions

TFIM on N spins, open boundaries, $H_S = -J \sum \sigma_i^x \sigma_{i+1}^x - h \sum \sigma_i^z$, $h > J$; coupling site $j_0 = \lfloor N/2 \rfloor$ (0-indexed), $S = \sigma_{j_0}^z$; thermal $\sigma = \gamma_S$ at $\beta \in \{0.5, 1, 2\}$.

Remark 5.1 (Site convention (v2 clarification)). For a distant probe site at separation ϵ we take $k = j_0 - \epsilon$ (equivalently, the maximum of the two ratios at $j_0 \pm \epsilon$; on an open chain with even N , $j_0 = N/2$ sits right of center and the two sides are *not* equivalent). v1 wrote $k = j_0 \pm \epsilon$ without fixing the branch; the tabulated values correspond to $k = j_0 - \epsilon$ and are reproduced exactly by the distributed script (Appendix E).

5.2 Witness ratios and results

$R(\epsilon; O) := \| [S(0), O] \|_{2,\sigma}^2 / \| O \|_{2,\sigma}^2$ and $R_{\text{opt}}(\epsilon) := \max_{O \in \text{span}\{\sigma_k^x, \sigma_k^y, \sigma_k^z\}} R(\epsilon; O)$; by Lemma 4.1, $\kappa(\epsilon) \geq \frac{\gamma(0)}{2} R_{\text{opt}}(\epsilon)$.

N	$R_{\text{opt}}(1)$	$R_{\text{opt}}(2)$	$R_{\text{opt}}(3)$
8	0.134648	0.0885503	0.0684289
10	0.0999406	0.0696450	0.0550748
12	0.0793451	0.0574518	0.0464596

Scaling tests with $\epsilon = \epsilon(N) \in \{\lfloor N/4 \rfloor, \lfloor N/3 \rfloor, \epsilon_{\max}(N), 3\}$ and temperature sweeps over $\beta \in \{0.5, 1, 2\}$ behave as in v1 (monotone decrease of $R_{\text{opt}}(\epsilon(N))$ over the tested sizes, non-negligible fixed- ϵ witnesses); the plots are regenerated by `verification/0073/witness_tfim.py` rather than shipped as loose image files (v1 Remark 1 removed).

6 State-side interface micro-tests

The weak-coherence families $\rho_E(\alpha) := \text{diag}(p) + \alpha \sqrt{p_m p_n}(|m\rangle\langle n| + |n\rangle\langle m|)$ (energy basis, $\Delta[\rho(\alpha)] = \text{diag}(p)$ by construction) give direct access to $r(\rho) := \dot{C}_{\text{loss}}(\rho)/C(\rho)$. As in v1 (unchanged; these tests use only exact semigroup steps and Lemma 4.1-type channels): for $N = 6$, $J = 1$, $h = 1.5$, $\beta = 1$, center coupling, $r(\rho)$ exhibits a plateau $\approx 0.157\text{--}0.158$ ($\{\pm\omega\}$ truncation) and $\approx 0.160\text{--}0.161$ with an added $\omega = 0$ channel at $\gamma(0) = 0.2$; in the $\omega = 0$ -only regime the plateau scales linearly in $\gamma(0)$ ($r/\gamma(0) = 0.2039$ across $\gamma_0 \in \{0.05, 0.1, 0.4\}$) and, at fixed pair $(m, n) = (0, 1)$ and fixed γ_0 , as $1/\beta$ (βr constant to 2×10^{-3} relative across $\beta \in \{0.5, 1, 2, 4\}$). Euler step-size and PSD-clipping stability checks as in v1 (plateau stable to 10^{-6} across $dt \in \{10^{-4}, 5 \times 10^{-5}, 10^{-5}\}$ and clipping thresholds $10^{-12}\text{--}10^{-16}$).

7 State–observable interface (explicit hypothesis)

The maintenance inequality bounds power via the state functional $\dot{C}_{\text{loss}}(\rho)$; $\kappa(\epsilon)$ is an observable-side ceiling. The bridge is stated as an explicit, falsifiable hypothesis, unchanged from v1: on families that are (1) weak-coherence, (2) population-nonextremal ($\Delta[\rho] \geq p_{\min} \mathbf{1}$), (3) coherence-supported in a fixed Bohr sector set Ω , and (4) within a validated Davies regime, one has $\dot{C}_{\text{loss}}(\rho) \geq (1 - \eta_\epsilon) \Gamma_\Omega C(\rho)$ with $\Gamma_\Omega := \min_{\omega \in \Omega} \lambda_{\min}(\Gamma(\omega))$ and $\eta_\epsilon \rightarrow 0$ as the coherence scale vanishes. The micro-tests of Section 6 are direct numerical probes of this interface.

8 Operational consequence (conditional, well-typed)

Combining Theorem 2.1(b) or (c) with the interface hypothesis: for ρ in such a family with $C(\rho) \geq C_0$,

$$P \geq k_B T (1 - \eta_\epsilon) \Gamma_\Omega C_0, \quad (7)$$

where P is total power under diagonal contraction, or incremental power over ϵ -efficient baselines (up to ϵ) under population autonomy.

9 Conclusion

We separated the corrected, imported thermodynamic maintenance bounds from finite-dimensional dynamical interface results. In Davies dynamics, the exact $\omega = 0$ identity yields operational witness lower bounds (reproduced exactly under the fixed site convention), and the corrected modular-weighted Bohr-block identity replaces v1’s false commutator decomposition — machine-verified, and reducing to the $\omega = 0$ case where all v1 numerics lived. Envelope suppression survives as a *linear* sufficient bound with explicit hypotheses; the quadratic Davies envelope is open, while being exactly realized in the quasi-free local-sink class [3]. TFIM witnesses ($N \leq 12$) remain finite-size diagnostics, not thermodynamic-limit claims.

A Numerical stability checks

As in v1: the plateau ratio $r = \dot{C}_{\text{loss}}/C$ moves by less than 2.3×10^{-6} across Euler steps $dt \in \{10^{-4}, 5 \times 10^{-5}, 10^{-5}\}$ and is bitwise stable across PSD clipping thresholds $\{10^{-12}, 10^{-14}, 10^{-16}\}$.

B Imported inequality: precise pointer map (v2)

Theorem 2.1(a) = [2] v2, Theorem on entropy production and rate-Pythagoras split; (b) = its total-power corollary under diagonal contraction; (c) = its efficient-baseline theorem and free-baseline proposition. v1’s pointer map targeted the v1 statement whose extra-power definition was vacuous (Remark 2.2).

C Reproducibility

All scripts formerly inlined in v1’s Appendix C are maintained, with reference logs, in the companion repository under `verification/0073: witness_tfim.py` (Table values, scaling and β sweeps, site convention of Remark 5.1), `verify_0073.py` (identity tests), and the micro-test scripts.

D Changes relative to version 1

1. **Lemma 2 corrected** (Lemma 4.3, Remark 4.4): v1’s Bohr-block commutator decomposition of the KMS Dirichlet form is false for $\omega \neq 0$ (measured 30–40% deviation at $\beta = 1$); the exact identity carries modular weights $e^{\beta\omega/2}$ and a non-commutator cross term, verified to 1.9×10^{-10} . Lemma 4.1 ($\omega = 0$) is the weight-1 special case and stands; all v1 witness numerics are unaffected.
2. **Lemma 3 rescope**d (Lemma 4.6, Remark 4.7): the quadratic envelope is withdrawn as proved (its proof used the false identity; the corrected splitting leaves T -linear cross terms and thermal-clustering residues, and the exact tail-reduction shortcut is numerically false, defect $\approx 5 \times 10^{-3}$); replaced by a linear-envelope sufficient lemma with explicit hypotheses, including a new thermal-clustering hypothesis (iv). The quadratic case is reposed as an open question, exactly realized in the quasi-free local-sink class [3].
3. **Imported theorem re-anchored** to [2] (v2) (Theorem 2.1, Remark 2.2); the no-go corollary retyped accordingly.
4. **Site convention fixed** (Remark 5.1): Table values correspond to $k = j_0 - \epsilon$; reproduced exactly by script.

5. **Editorial:** v1’s compile-time figure-file remark removed (artifacts script-generated); program map updated to the v2 series (P1–P7 with corrected descriptions); inline Colab listings moved to the repository; ceiling/floor annotation added to $\kappa(\epsilon)$ per [4] (v2).

E Verification summary

Distributed scripts (TFIM $N = 8$, $\beta = 1$, $h = 1.5$, $J = 1$, exact Bohr blocks): (i) corrected identity (2) vs. the direct Dirichlet form: worst relative deviation 1.9×10^{-10} over random Hermitian O (positive-frequency blocks); v1’s commutator form: deviations 0.30–0.39. (ii) Witness table reproduced exactly under Remark 5.1 (0.134648, 0.0885503, 0.0684289 at $\epsilon = 1, 2, 3$, $N = 8$); the $k = j_0 + \epsilon$ branch gives 0.088550, 0.068429, 0.053108, quantifying the open-chain asymmetry. (iii) Exact tail-reduction defect: $\mathcal{E}(O; S)$ vs. $\mathcal{E}(O; \text{tail } S)$ differ by 4.6×10^{-3} relative at $\epsilon = 3$ for a far-supported two-site O , confirming Remark 4.7.

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